



A Systematic Review of Conversational AI Chatbots in Academic Advising

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Abstract. Purpose – This paper aims to review several studies published between 2018 to 2022 about advising chatbots in schools and universities as well as evaluating the state-of-the-art machine learning models that are deployed into these models.

Methodology – This paper follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), it demonstrated the main phases of the systematic review, it starts with screening 128 articles and then including 11 articles for systematic review which focused on the current services of the advising chatbots in schools and universities, as well the artificial models that are embedded into the chatbots.

Findings – Two main dimensions with other sub-dimensions are extracted from the 11 included studies as it shows the following: 1- Advising chatbots AI Architecture which includes other sub-dimensions on identifying the deep learning based chatbots, hybrid chatbots and other open-resources for customizing chatbots; 2- The goals of the advising chatbot as it includes both the admission advising and academic advising.

Conclusion – Most of studies shows that advising chatbots are developed for admission and academic advising. Few researchers who study the chatbots in high schools, there is a lack of studies in developing chatbots for students advising in high schools.

Limitations and future work – This study is constrained to review the studies from 2018–2022, and it is not exposed to the chatbots artifacts, even though, the human-chatbot interaction has an essential impact on students' experiences. Future research should include the impact of chatbots interactive design and students' experiences.

Keywords: Artificial Intelligence · Chatbot · Conversational AI · Machine Learning · Student · Systematic Literature Review

1 Introduction

Artificial Intelligence plays an essential role in developing a chatbot that can learn how to understand human and making a decision. The most important components of the artificial intelligence that can make a successful chatbot are the natural language processing

(NLP) and machine Learning (ML) (Khan & Rabbani, 2021; Bird et al., 2021). In general, the chatbot works by taking the natural language instructions as input and the output is expected to be very relevant to the user's input (Caldarini et al., 2022). Interestingly this is explained the reason behind that machine learning is becoming increasingly ubiquitous across different industries including education, finance, healthcare, etc. Hence, the researchers and developers around the world are inspired to develop the state-of-the-art chatbots by embedding different machine learning algorithms such as naïve Bayes algorithm and support vector machine (SVM) (Bird et al., 2021; Tamizharasi et al., 2020; Kushwaha & Kar, 2020; Assayed et al. 2022) and other deep learning algorithms mainly the convolutional neural network (CNN), long short term memory network (LSTM) and recurrent neural network (RNN) (Dhyani et al., 2021; Sperlí, 2021; Prasetyo & Santoso, 2021; Kliestik et al., 2022; Assayed et al., 2023). Accordingly, several authors conducted different reviews about state-of-the-art chatbots in education and learning. Ji et al. (2023) conducted a review about learning classes and subjects in schools without comparing the state-of-the-art chatbots technology and applications. On the other hand, Nee et al. (2023) conducted another systematic review to define the potential contribution between educational tools in elementary, middle, high school and universities, with no details about the models or algorithms that are deployed in these chatbots. Furthermore, Assiri et al. (2020) proposed a review of academic advising computer systems in undergraduate education without focusing on chatbots. Despite the importance of readers and developers to evaluate the architecture as well as the performance of advising chatbots but few authors who defined the novelty in chatbots applications. For instance, developers need to know the number layers that used in deep learning model, the type of machine learning that deployed into the system, and etc. Therefore, this study aims to close this gap by conducting a systematic literature review (SLR) to propose the students advising chatbots for particularly students in high schools or prospective students in higher education. Accordingly, this paper reviewed several studies by considering chatbots for students advising and evaluating the AI and machine learning models that are deployed into these agents. Three research questions will be formulated in this paper:

RQ1: What are the current AI models and algorithms that deployed into the Advising chatbots?

RQ2: What are the current Advising chatbots services at universities?

RQ3: How do AI conversational advisers support high school students?

The SLR in this paper will provide numerous state-of-the-art academic advising chatbots' architectures as well as exposing the developers and researchers to the main goals and services that can be used to enhance the students' services into different educational institutions. This study organized as follows: Sect. 2 explains the methodology and the approach that adopted in this paper, while Sect. 3 presents the discussions and results, finally the conclusion and limitation are both presented in Sect. 4.

2 Methodology

2.1 Approach

This paper follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), it demonstrated the main phases of the systematic review, it identifies the number of study articles that are explored, identifying the excluded and included articles with stating the reasons. The main advantage of using PRIMSA is to improve the transparency for this review and providing a clear analysis for all the eligible study articles (Selçuk, 2019). Figure 1 illustrates the entire approach and it shows more details in each single stage.

2.2 Resources

The below Table 1 and Table 2 show the databases & journals that were used in exploring the articles and other references.

Table 1. The Databases and Journals that used in this study

Database	Number of Studies
WorldCat.org	87
Academic Search Complete	10
Computers & Applied Sciences Complete	12
ScienceDirect	2
ProQuest Central	13
ABI/INFORM Global	4
Electronic Books (27)	27
SAGE Journals (1)	1
Total	156

Table 2. Google Scholar search criteria

Google Scholar	6
All Articles	162

2.3 Search Criteria

The search strategy conducted by using the resources that are mentioned in previous Sect. 2.2. The search implemented by using Boolean operators (AND, OR). The criteria include three categories and it's defined in three statements as explained in Table 3. However, in this study the authors adopted the search criteria to retrieve publications between 2018–2022, in order to study the state-of-the-art advising chatbots.

Table 3. The search categories and criterias.

code	Search Category	Criteria
A	Chatbot	(chatbot OR dialogue OR “conversational AI” OR “conversational agent” OR “digital assistant” OR “Virtual Assistant”)
B	AI-based Approach	(“Machine Learning” OR “Deep Learning” OR “Neural network” OR “Artificial Intelligence”)
C	Education	(education OR academic OR adviser OR advisor OR advising OR school OR “high school” or “college” OR “university” OR “Universities” OR “students” OR “students”)
	Formula=	“A” AND “B” AND “C” ti:(chatbot OR dialogue OR “conversational AI” OR “conversational agent” OR “digital assistant” OR “Virtual Assistant”) AND ti:(“Machine Learning” OR “Deep Learning” OR “Neural network” OR “Artificial Intelligence”) AND ti:(education OR academic OR adviser OR advisor OR advising OR school OR “high school” or “college” OR “university” OR “Universities” OR “students”)

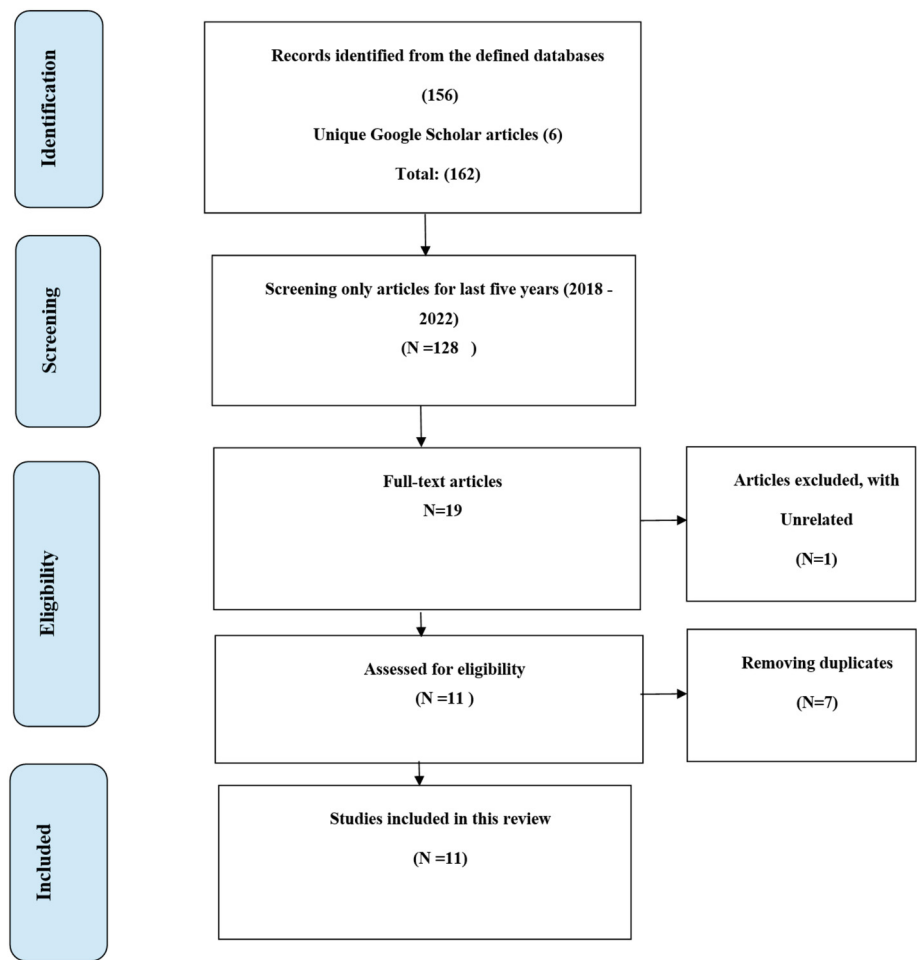


Fig. 1. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). It demonstrates the main phases of the systematic review.

2.4 Identification

The articles are collected based on the defined criteria in Sect. 2.3 and the number of articles that are retrieved from the databases that mentioned in Sect. 2.2 are (N = 162).

2.5 Screening and Eligibility

The number of records are limited to the five years (2018–2022) with total number (n = 128), afterward we select only the full-text documents, with total number is (N = 19).

2.6 Excluded Strategy

Some documents are excluded from the screening list, the article from Suárez et al. (2022) is excluded from the list, the title met all the predefined search criteria the title: “Using a Virtual Patient via an Artificial Intelligence chatbot to Develop Dental Students’ Diagnostic Skills”, the context is about the dental’s students satisfactions with AI chatbot that enhance interacting with virtual patients.

2.7 Included Strategy

The number of studies that are considered in the literature review are ($N = 11$), these articles are all about using the chatbot or conversational AI in assisting and supporting students at universities, students are the targeted audience in all studies. Table 4 shows these studies.

2.8 Literature Taxonomy

According to Randolph (2009) the effective literature review should begin with matching the proposed thesis’s review with cooper’s taxonomy of literature reviews. Cooper’s (1985) proposed to have a taxonomy for the scope of literature review which includes the following: The focus of the research, the goal, the perspective of the reviewers, coverage, organization and the audience. According to our selected (11) studies, we can construct the taxonomy of the literature review from the authors perspectives toward the main topics of their articles. The taxonomy in this literature is constructed into two main categories, the first one is academic chatbot’s AI architecture, the second category is the goals of academic chatbots and the. Each particular category explored the contributions of students advising and counseling. However, the majority of the studies are focused on the deep learning algorithms and basically in seq2seq architecture such as BRNN, BLSTM, and attention models in benefiting particularly the admission department at universities. Figure 2 illustrates the taxonomy of the selected articles.

3 Results and Discussion

This reviews investigates the state-of the-art advising chatbots from 2018 to 2022 and two main dimensions with other sub-dimensions are extracted from the 11 included studies as it shows the following: 1- Advising chatbots AI Architecture which includes other sub-dimensions on identifying the deep learning based chatbots, hybrid chatbots and other open-resources for customizing chatbots; 2- The goals of the advising chatbot as it includes both the admission advising and academic advising.

The below findings are answered to the research questions that are highlighted in the introduction section.

Table 4. The included studies in the systematic review

Article Name	Author	Result	Goal	Evaluation/Dev	Academic Service	Target audience	AI-Approach
Artificial Intelligence Based University Chatbot using Machine Learning	Khan et al. (2021)	The results show that the Random Forest is outperform for the proposed study	Supporting people which are visiting the admission office this will also decrease the traffic of the admission office	Evaluating the Models	University Admission	Parents and Prospected students at universities	Supervised Machine Learning (DT, RF, SVM)
Implementation of a Virtual Assistant for the Academic Management of a University with the Use of Artificial Intelligence	Villegas-Ch et al. (2021)	The results show an increasing number of the enrolled students in the training program since students they have more knowledge about the program	Providing more information on an academic courses at universities and reducing the administrative budgets	Evaluated by using the survey to the training programs students	Admission section-training programs offered by the university	College students	Language Understanding (LUIS) is a cloud-based conversational AI
Engaging Students With a Chatbot-Based Academic Advising System	Kuhail et al. (2022)		that helps students with prescriptive academic inquiries		Academic Advising	Current university students	
A Chatbot to Support Basic Students Questions	Santana et al. (2021)	Delivering a high accuracy in the classification of enquiries intention The chatbot is able to answer the most FAQ	Helping new students in the general administrative procedures and processes	Developing a new model (chatbot)	Academic Advising-	Freshmen students	NLP and Machine Learning
Artificial Intelligence based Chatbot for Placement Activity at College Using DialogFlow	Ranavare & Kamath (2020)	This Chatbot is integrated to institute's website by clicking the Integrations choice in the left panel to generate a web demo for present agent and then press the Web Demo tile button	This agent provides information about placement activities to students	Developing a model/Integration	Internship (placement activities) service	Current students in universities (Internship)	DialogFlow and Natural Language Processing (NLP)
Jooka: A Bilingual Chatbot for University Admission	El Hefny et al. (2021)	The System Usability Scale (SUS) and Chatbot Usability Questionn (CUQ) were used to evaluate it terms of usability acceptable value is <62.6"	A bilingual chatbot for supporting students and German University in Cairo (GUC)' staff to enhance their admission process	Developing a chatbot	University Admission	Prospected Students (High School)	(Dialogflow ES) and Amazon Web Services (AWS)

(continued)

Table 4. (continued)

Article Name	Author	Result	Goal	Evaluation/Dev	Academic Service	Target audience	AI-Approach
College Enquiry Chatbot using Rasa Framework	Meshram et al. (2021)	It's done by having a confusion matrix and performance measures like Precision, Accuracy & F1 Score and the results 0.628, 0.725 and 0.669	Answering students queries about any admissions-related question	Developing a chatbot	University Admission	New students at Universities and prospecting Students	Natural Language Understanding with RSA NLU-Open Resource Technology
CollegeBot: A Conversational AI Approach to Help Students Navigate College	Daswani et al. (2020)	The semantic (similarity mode) outperforms seq2seq model even though the seq2seq model is faster than semantic similarity	To assist visitors of a university's web site to locate answers to their queries	Evaluating two algorithms for training Chatbot & model proof-of-concept prototype to demonstrate adopted by other colleges	Students orientations	Prospected and current students (All university's website visitors)	NLP (Semantic Similarity) and RNN-based Sequence-to-Sequence (seq2seq)
Bilingual AI-Driven Chatbot for Academic Advising	Bilquise et al. (2022)	The chatbot engine determines the user's intent by processing the input and retrieving the most correct response that matches the intent with an accuracy of 80% in English and 75% in Arabic	Supporting current students in universities by answering to their academic questions such as the requirement courses, electives and GPA	Developing a bilingual chatbot (Arabic/ English)	Academic Advising-College students	Current students in Colleges	Neural network and Natural Language (NLP) technologies
Indonesian chatbot of university admission using a question answering system based on sequence-to-sequence model	Chandra, Y. W., & Suyanto, S. (2019)	Using the Metric of BLUE score: 44.6	Customer service-Admission office by integrating it with Whatsapp instant messaging	Developing a chatbot in Telekom University in Indonesia	Students Admission	New Students	Seq2Seq-bidirectional LSTM with Attention in the Encoder-Decoder-
Developing a Chatbot system using Deep Learning based for Universities consultancy	Le-Tien et al. (2022)	The results achieve 89% F1-score of 3 classes of the Text Classification in Deep Learning	Advising the current students toward their academic plan	A Proof of Concept to the Ho Chi Minh City University of Technology (HC-MUT)	Academic Advising	Current Students	Bidirectional-LSTM with Attention mechanism

3.1 Advising Chatbots AI Architecture

This section answers the below research question:

RQ1: What are the current AI models and algorithms that deployed into the Advising chatbot?

Haristiani (2019) views the chatbot as one the advanced technologies that can be applied in education, inspiring researchers to explore different aspects of chatbots. The performance and the efficiencies of the AI chabots depend basically on the state-of-the-art algorithms that are selected to use it (Santana et al., 2021). There are multiple types

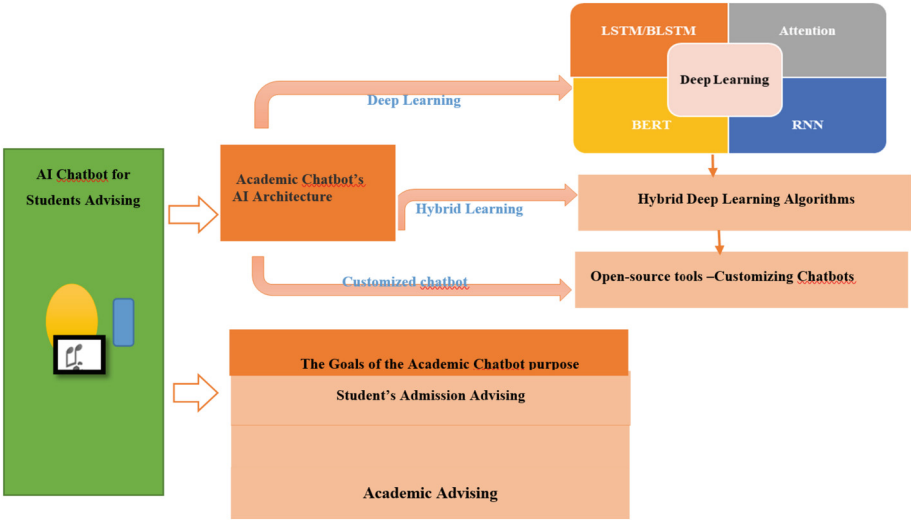


Fig. 2. The Taxonomy of the Selected Articles

of machine learning that are used in the models, However, Fig. 3 depicts the revealed studies based on AI architecture.

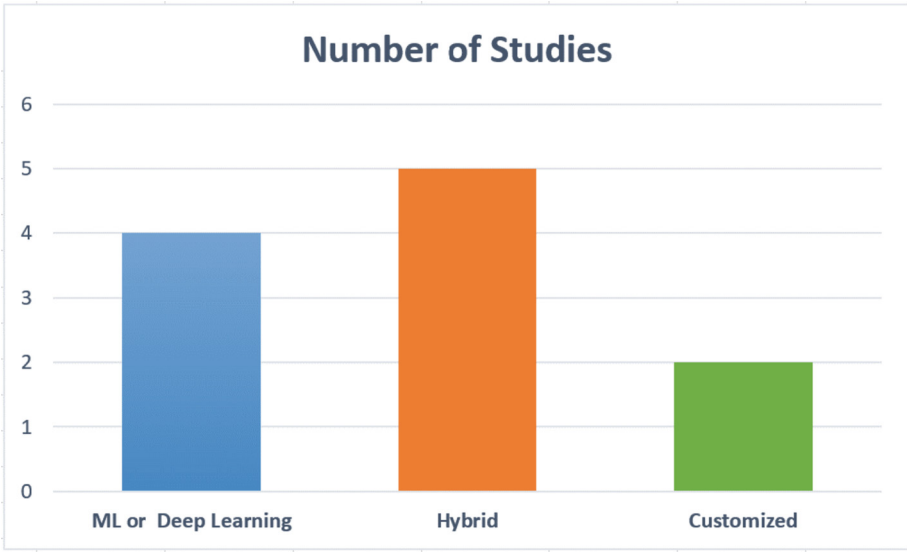


Fig. 3. The AI based chatbots architecture

3.1.1 Deep Learning

Khan et al. (2021) proposed the random forest algorithm, decision tree and support vector machine for supporting students and parents who visit the campus. The authors started with preprocessing the dataset by using the features engineering such as TFIDF with adding the Ngram feature and the results shows that Decision Tree algorithms and Support Vector Machine are not responding well like Random Forest algorithm, thus, the RF model is adopted in the system as an effective approach for assisting students in universities. In fact, the result of any model depends on the validation that are accepted. Recently many developers are inspired in using the seq-2seq model by adding more layers such as attention layers and LSTM/ BLSTM. Chandra & Suyanto (2019) developed a chatbot in one of the Indonesian universities by using a small dataset which collected from the admission office. The system developed by using seq2seq model, since the data is basically a conversation between the admission officers and the students, however, the authors added BLSTM layers as well as attention layer in order to improve the accuracy of the chatbot. Furthermore, another deep learning chatbot is implemented as a proof of Concept in Ho Chi Minh City University of Technology (HC-MUT) by combining the Embedding layer and the Bidirectional-LSTM with the Attention mechanism for solving the classification tasks from students' majors-degree requests (Le-Tien et al., 2022).

3.1.2 Hybrid Learning

There are several authors conducted studies in the hybrid learning, which can vary in the complexities of the model, some models might have a combination between the supervised machine learning and deep learning, and others might have a combination between different deep learning models together. For example, Yu et al. (2020) developed a hybrid model for wearable inertial sensor-based systems in order to prevent older people falls that can cause them injuries, this model called ConvLSTM which outperformed the both CNN and LSTM individually. However, In reference to the literature in this study, a few studies focused on the hybrid learning, Chandra & Suyanto (2019) developed a chatbot as customer service in the Telekom University in Indonesia for assisting students in the admission processes, the authors adopted the Seq2Seq approach with combining both the BI-LSTM and Attention models together.

3.1.3 Open-Source Tools for Customizing AI Chatbots

Some authors design the chatbots by using AI packages such as Rasa Natural Language Understanding (NLU) which is an open source framework for NLP solution that take user's input and try to extract the intent, Meshram et al. (2021) adopted this configuration in the proposed chatbot, after the data is trained, they fed it into the NLU model pipeline, however they used the Anaconda platform and python language in order to deploy the required packages.

3.2 The Goals of the Advising Chatbots.

In this section the below research questions are answered:

RQ2: What are the current Advising chatbots services at universities?

RQ3: How do AI conversational advisers support high school students?

Several researchers studied the dialogue systems and advising chatbots in schools and universities. However, few of them who investigated the advising chatbots in high schools. Assayed et al. [23] developed a machine learning chatbot called “HSchatbot” to assist students in some of high schools to classify their enquiries based on type of their questions, nevertheless, most of the authors focused on studying the impacts of chatbots on students’ admission at universities. Since most universities and colleges around the world receive thousands of applications yearly during a limited time and as a result the admission officers would receive many calls from students and parents (Fig. 4).

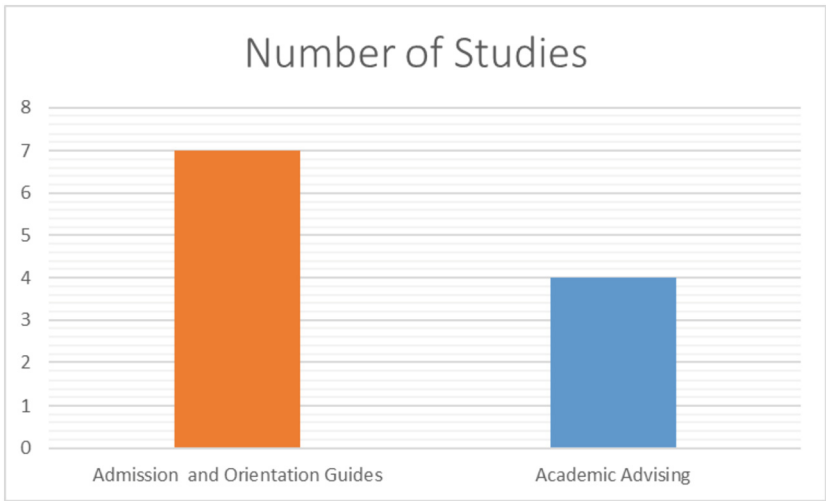


Fig. 4. The distribution of academic chatbots based on goals

3.2.1 Students Admission Advising

El Hefny et al. (2021) developed a chatbot called “Jooka” for supporting prospected students who are targeting the German University in Cairo (GUC), which can answer to their admission enquiries. Jooka enhances the advising process by providing instant responses to the both students and parents in both languages (English and Arabic). Moreover, Khan et al. (2021) proposed a model for supporting students and parents to provider in 24/7 services by answering any question that are related to registration and admission. Another chatbot called “CollegeBot” that developed by Daswani et al. (2020) to assist the university’s website’s visitors to navigate the website effectively, however, the authors in this study present a proof-of-concept model for the CollegeBot to the San Jose State University. Moreover, Meshram et al. (2021) developed a chatbot called “college enquiry chatbot” that aims to answer any admission-related questions.

3.2.2 Academic Advising

Students-faculty advising play a vital role on students' success (Sneyers, & De Witte, 2018), advisers can assist the students in making the right decision in selecting the courses as well as monitoring the progress of students in order to be sure to graduate on-time. Bilquise et al. (2022) developed a bilingual chatbot for supporting the current students academically in order to be sure that are following the academic plan as well as to graduate on-time and it supports both languages the Arabic and English. Another novel chatbot is developed by Le-Tien et al. (2022) for advising the current students in Vietnamese universities to select the appropriate courses that match with their academic plan and accordingly students would be able to declare their major successfully.

4 Conclusion

This review investigates the current state-of-the-art advising chatbots in schools and universities. Most of studies shows that advising chatbots are developed for admission and universities academic advising. Despite the vital role of academic advisers in high schools where students' future careers are shaped by choosing the best fit majors and universities, few studies conducted on chatbots that focused on this stage, resulting in the gap in the development chatbots for student advising in high schools.

4.1 Limitations and Future Work

This study is constrained to review the studies from 2018–2022 utilizing specific accessible databases, moreover, the taxonomy used in this SLR does not cover the chatbots artifacts, even though, the human-chatbot interaction have an essential impact on students' experience. Future research should include the impact of chatbots design and students' experiences while also exposing to other familiar databases such as Web of Science and Scopus.

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